GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-ROBO-LINK-150_RevA.STEP) GOVERNS NOMINAL GEOMETRY FOR ALL UNTOLERANCED SURFACES.
   1.3 GENERAL TOLERANCES: LINEAR +/- 0.05 mm, ANGULAR +/- 0.2 DEG (PER ISO 2768-m).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS ARE IN MILLIMETERS.

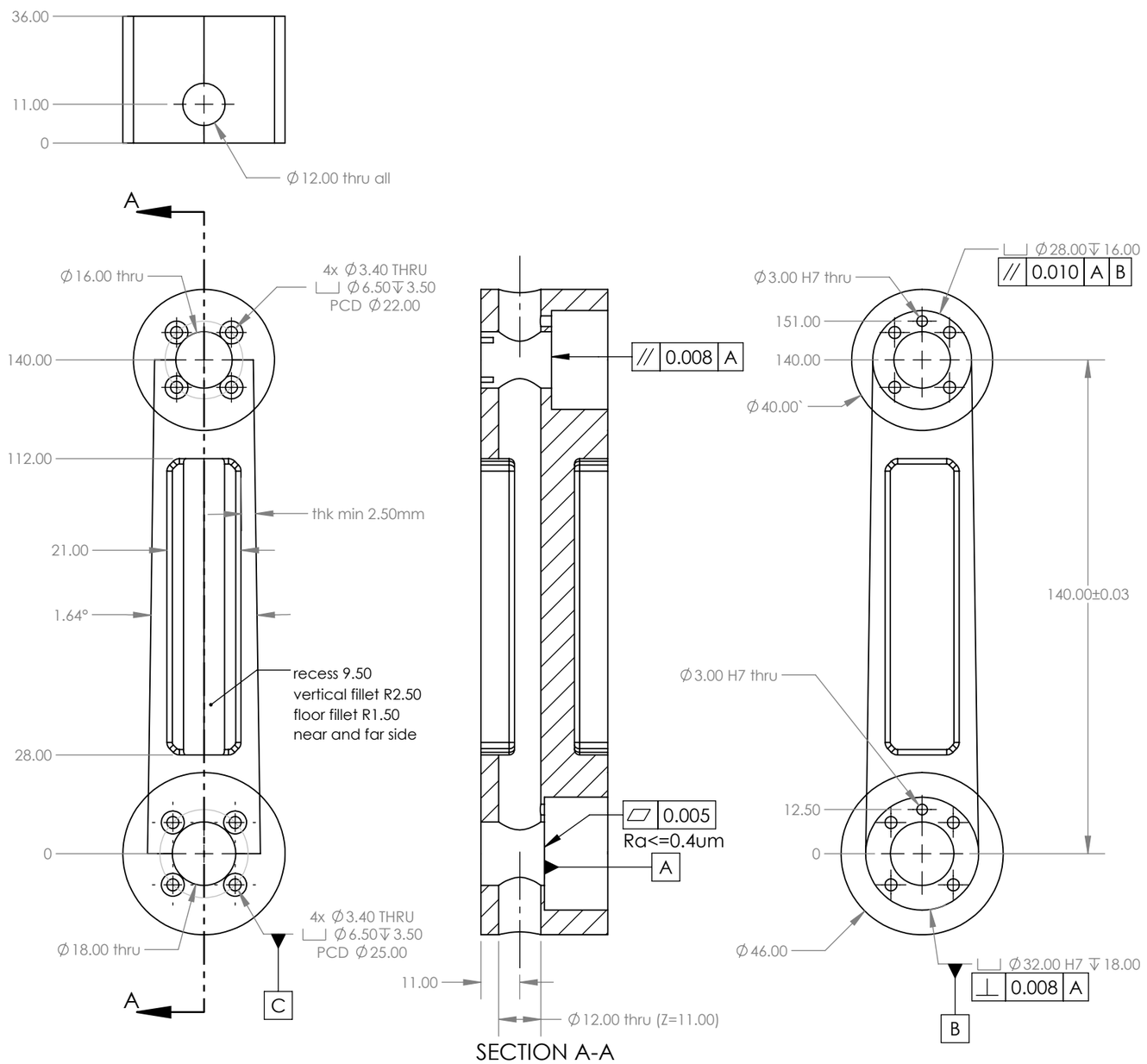
2. MATERIAL & BLANK PREPARATION:
   2.1 MATERIAL: HIGH-STRENGTH PRE-STRETCHED ALUMINUM ALLOY 7075-T651 PER ASTM B209 / AMS 4045.
   2.2 MECHANICAL PROPERTIES (LONGITUDINAL):
      - YIELD STRENGTH: Rp0.2 >= 505 MPa.
      - TENSILE STRENGTH: Rm >= 570 MPa.
      - MODULUS OF ELASTICITY: E = 71.7 GPa, DENSITY = 2.81 g/cm^3.
   2.3 THEORETICAL CALCULATED FINISHED MASS: 135.0 g (+/- 3.0 g).
   2.4 STRESS RELIEF: ULTRASONICALLY STABILIZED PRIOR TO HIGH-SPEED FINISH MACHINING TO PREVENT BOWING DISTORTION.

3. MANUFACTURING & CNC DFM CONTROLS:
   3.1 100% SUBTRACTIVE HIGH-SPEED CNC MILLING (SPINDLE SPEED >= 18,000 RPM, ZERO LATERAL UNDERCUTS).
   3.2 POCKET CORNER RADIUS: ALL VERTICAL INTERNAL CORNERS SHALL BE R 2.50 +/- 0.10 mm (SUITABLE FOR DIA 4-5 mm ENDMILL).
   3.3 POCKET FLOOR TRANSITION RADIUS: R 1.50 +/- 0.15 mm MANDATORY (BULL-NOSE ENDMILL PROFILE).
   3.4 STRICT WALL THICKNESS CONTROL: MINIMUM STRUCTURAL WALL THICKNESS SHALL NOT BE LESS THAN 2.50 mm AT ANY POINT.
   3.5 CENTRAL HARNESS CONDUIT (DIA 12.0 mm THRU) MUST REMAIN COMPLETELY ISOLATED WITH ZERO BREACH INTO TOP/BOTTOM POCKETS.

4. PRECISION INTERFACES & FINISH MACHINING:
   4.1 PROXIMAL BEARING SEAT: DIA 32.000 +0.012/-0.000 mm (H7), CYLINDRICITY <= 0.005 mm, Ra <= 0.4 MICRONS.
   4.2 DISTAL BEARING / HARMONIC SEAT: DIA 28.000 +0.012/-0.000 mm (H7), CYLINDRICITY <= 0.005 mm, Ra <= 0.4 MICRONS.
   4.3 DATUM A (PROXIMAL BASE FLANGE) FLATNESS <= 0.005 mm, Ra <= 0.4 MICRONS.
   4.4 8X DIA 3.40 mm THRU WITH BOTTOM COUNTERBORES DIA 6.50 mm X 3.50 mm DP ON PCD 25.00 / 22.00 mm.

5. SURFACE TREATMENT & CLEANROOM COMPLIANCE:
   5.1 SURFACE HARDENING: TYPE III HARD ANODIZE COATING PER MIL-A-8625 (CLASS 1, NON-DYED NATURAL).
   5.2 ANODIC LAYER THICKNESS: 25 TO 30 MICRONS (MICRO-HARDNESS >= 450 HV).
   5.3 MASKING DIRECTIVE: 100% MASKING REQUIRED ON DIA 32.000 H7, DIA 28.000 H7 BORES AND DATUM A FACE TO PREVENT TOLERANCE DRIFT.
   5.4 CLEANROOM RATING: COMPLIANT WITH ISO 14644-1 CLASS 3 AIRFLOW. ZERO PARTICULATE FLAKING OR OIL RESIDUE PERMITTED.
   5.5 FINAL DECONTAMINATION: MULTI-STAGE ULTRASONIC CAVITATION IN DEIONIZED WATER (18 MOhm*cm); VACUUM BAKED AT 80°C FOR 2 HOURS.

6. DYNAMIC PERFORMANCE & QUALITY ACCEPTANCE (FAIR):
   6.1 100% CMM AUDIT ON BEARING BORE CENTER-TO-CENTER SPAN (140.00 +/- 0.03 mm) AND PARALLELISM.
   6.2 DYNAMIC INTEGRITY: FIRST NATURAL FREQUENCY f1 >= 2.1 kHz; 40G ACCELERATION RESIDUAL SETTLING TIME <= 3.5 ms.



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH: MIL-A-8625 Type III Hard Anodize (Masked Bores)		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE		TITLE:	
DRAWN					HUMANOID ROBOT ARM HIGH-DYNAMIC LIGHTWEIGHT LINK	
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Aluminum Alloy 7075-T651 (ASTM B209 / AMS 4045)	DWG NO. W3-ROBO-LINK-150_RevA	A3
				WEIGHT: 135.0g	SCALE: NTS	SHEET 1 OF 1